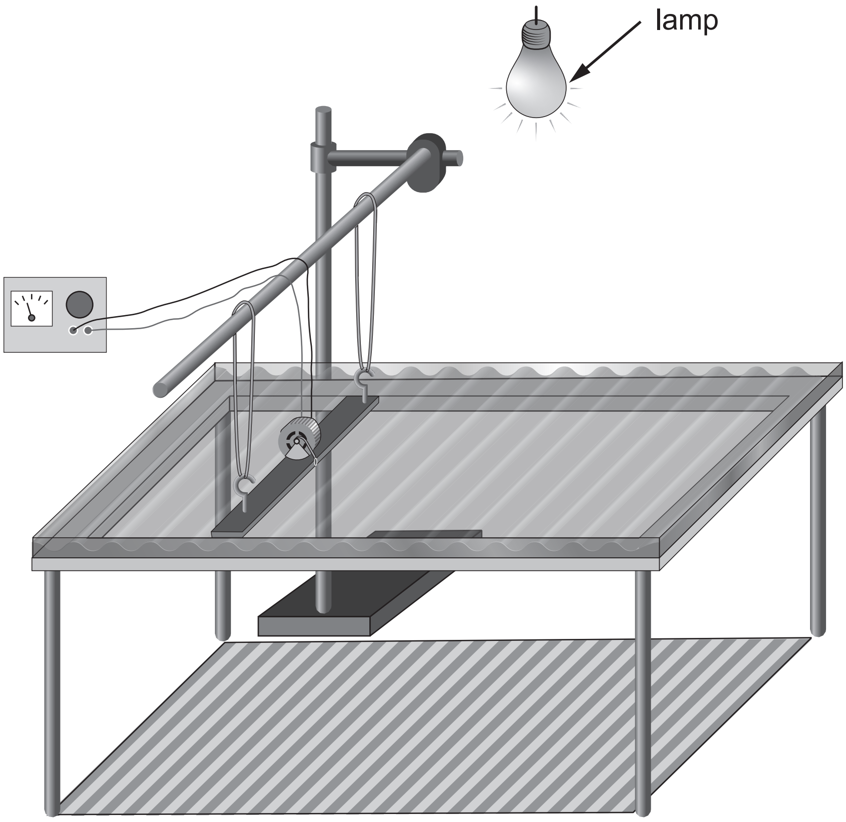
2. A teacher demonstrates waves using a ripple tank.
She changes the frequency of the waves produced and the class observes the effect on their wavelength.



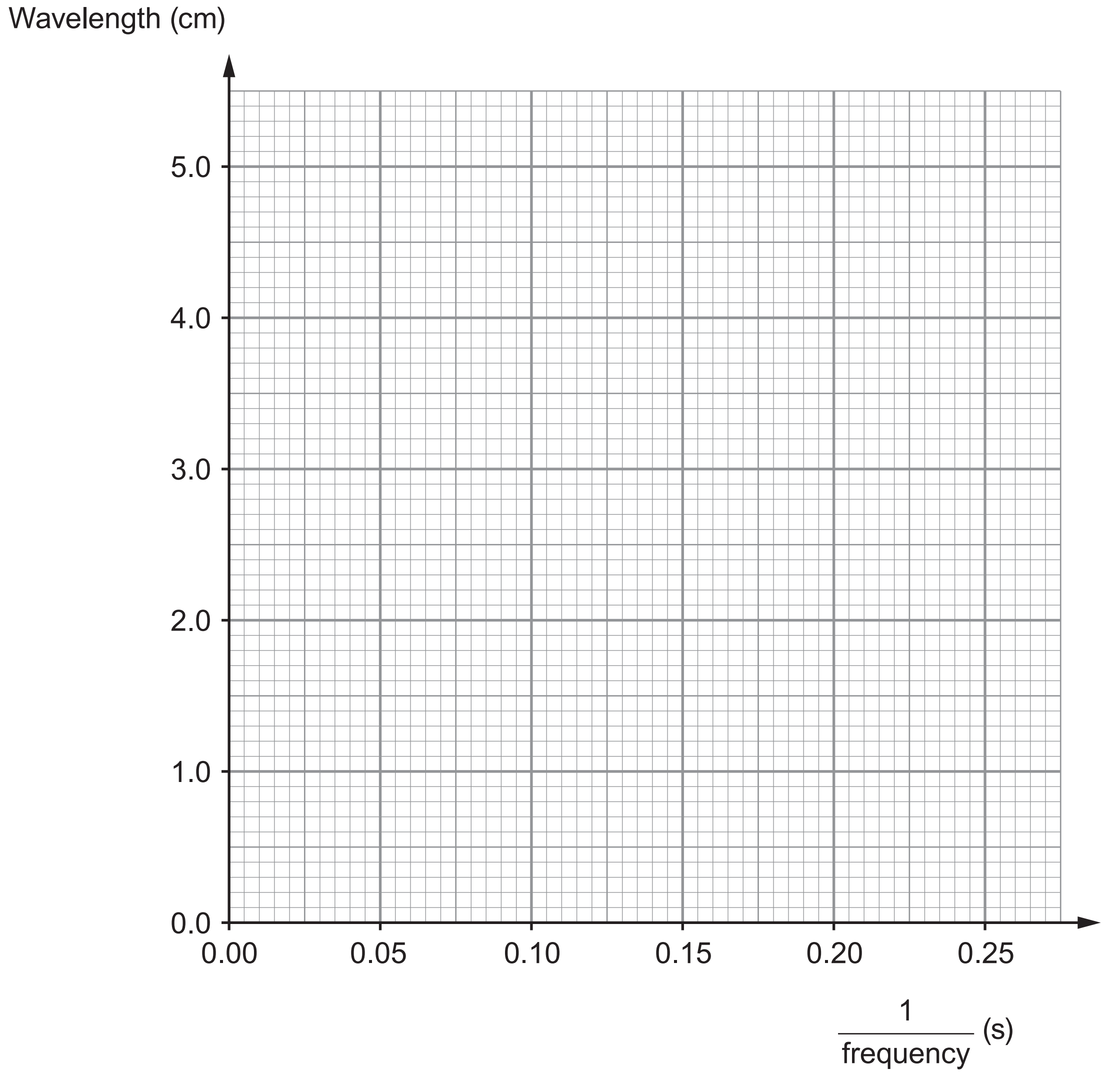
Next, the class investigates the link between frequency and wavelength using a virtual ripple tank simulation.

The table below shows their results.

Frequency (Hz)	$\frac{1}{\text{frequency}}$ (s)	Wavelength (cm)
20.0	0.05	1.0
10.0	0.10	2.0
6.7	0.15	3.0
5.0	0.20	4.0
4.0	0.25	5.0

Examiner only

(a) (i) Plot the data on the grid below and draw a suitable straight line. [3]



(ii) **Complete** the following sentences by underlining the correct word or phrase in the brackets. [2]

I. As $\frac{1}{\text{frequency}}$ increases the wavelength **[increases / decreases / stays the same]**.

II. As frequency increases the wavelength **[increases / decreases / stays the same]**.

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(iii) I. Use data from the table to state the wavelength of the waves at a frequency of **10 Hz**. [1]

Wavelength = ... cm

II. Use the equation:

wave speed = frequency × wavelength

to determine the speed of the waves at a frequency of **10 Hz**. [2]

Speed = ... cm/s

(b) (i) Electromagnetic waves are used to communicate with satellites. Some satellites remain above the same point on the Earth to allow constant communication.

Complete the following sentences about communications satellites by <u>underlining</u> the correct word or phrase in the brackets. [4]

Electromagnetic waves are **(longitudinal / parallel / transverse)** waves.

TV signals are sent to satellites in **(geothermal / geosynchronous /**

geostationary) orbits using **(microwaves / visible light / gamma rays)**.

These satellites orbit above the **(poles / equator / axis)** of the Earth.

(ii) A satellite orbits the Earth in a circular orbit, once every 24 hours. The radius of its orbit is 42 164 km.

I. Use the equation:

circumference of a circle = $2\pi r$ (where r = radius and π = 3.14)

to calculate the distance the satellite travels in one orbit. [1]

Distance = ... km

Examiner only

II. Use the equation:

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

to calculate the speed of the satellite in km/h. [2]

Speed = ... km/h

(iii) Maddie suggests that the satellite orbits at the same speed as a point on the Earth's surface moves, so that it always stays above the same point on the Earth. Explain whether or not you agree. [2]

....

....

....

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